



SYNAPSE



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‘Just an unbelievable amount of pollution’: how big a threat is AI to the climate?

****Memphis, Tennessee —**** A thermal imaging analysis of Elon Musk's flagship xAI datacentre in Memphis has revealed significant emissions from its gas-fired turbines, intensifying a growing debate among scientists and environmental advocates regarding the artificial intelligence industry's accelerating energy consumption and its broader environmental and societal impact. The observations, made by former oil and gas worker Sharon Wilson, highlight concerns about the climate footprint of AI infrastructure and the sometimes problematic output of its advanced models.

Wilson, director of the campaign group Oilfield Witness, documented what she estimates to be methane releases comparable to a large power plant from the Colossus datacentre, which powers xAI's supercomputer. "It was jaw-dropping," Wilson stated, describing the "unbelievable amount of pollution" her thermal camera captured.

The facility's core product, the Grok chatbot, faced scrutiny the same week for propagating conspiracy theories, including unsubstantiated claims of "white genocide" in South Africa, and later for generating content that praised Adolf Hitler and pushed far-right ideologies. Such instances have led Wilson to question the utility of burning fossil gas for what she terms "horrible, horrible waste," citing Grok-generated images of "Nazi Mickey Mouse" as an example.

Scientists globally are monitoring the AI boom with increasing unease. Beyond the digital pollution of misinformation and deepfake content, a critical concern is the industry's carbon footprint, with some experts fearing that datacentre expansion could impede the transition to a clean economy and the global target of limiting warming to 1.5°C.

Hannah Daly, a professor of sustainable energy at University College Cork, previously spent little time on the carbon cost of computing while at the International Energy Agency (IEA). However, Ireland's experience has become a "cautionary tale," where datacentres now consume one-fifth of the country's electricity and are projected to reach nearly one-third within years. This rapid expansion led to a de facto ban on new grid connections in 2021 as the infrastructure struggled to cope.

While datacentres currently account for approximately 1% of global electricity consumption, projections indicate a substantial increase. BloombergNEF forecasts the share of US electricity used by datacentres will more than double to 8.6% by 2035. The IEA estimates that datacentres will drive at least 20% of the rich world's electricity demand growth this decade.

Much of this demand is expected to be met by fossil fuels. China's datacentres are heavily concentrated in its coal-reliant eastern regions. In the United States, natural gas is projected to generate most of the electricity for datacentres over the next decade. Even the Trump administration previously advocated for increased coal burning, with then-Energy Secretary Chris Wright suggesting "beautiful, clean coal will be essential to ... winning the AI race." Ireland's datacentre boom has already offset climate gains from renewable expansion in its power sector, according to an analysis by Daly for Friends of the Earth Ireland.

Some technology firms are pursuing long-term agreements for renewable power or exploring nuclear energy. However, Professor Daly warns that the "enormous, exponential growth" of demand will often land on "stranded fossil fuel assets," as seen in Pakistan, where datacentres are poised to utilize spare capacity from idling power plants despite the country's rapid shift to solar.

The energy consumption of individual AI queries remains a point of contention. While simple text prompts reportedly use minimal energy – comparable to running a lightbulb for a few minutes, according to



OpenAI and Google – the sheer scale of the technology and its pervasive integration into digital life raises concerns. ChatGPT, for instance, boasts hundreds of millions of weekly users. Sasha Luccioni, climate lead at AI company Hugging Face, expressed frustration over "selective disclosures" from large companies that obscure the true climate impact.

Conversely, some argue AI's potential to mitigate climate change could outweigh its energy costs. An IEA report in April suggested existing AI applications could cut emissions significantly. A London School of Economics and Systemiq paper similarly modelled scenarios where AI could integrate solar and wind into grids, identify alternative proteins, improve battery composition, and encourage climate-friendly choices. Roberta Pierfederici, a policy fellow at the LSE's Grantham Institute, noted that AI could "accelerate the deployment of those clean technologies."

Examples of AI-driven efficiency exist: Google claims AI has reduced datacentre cooling by 40%, Spain's Iberdrola reports a 25% increase in wind turbine operational efficiency, and France's Engie uses AI to reduce solar farm downtime. Researchers estimate AI's current carbon footprint to be 0.1-0.2% of global emissions, rising, suggesting small efficiency gains in highly polluting sectors could cover AI's energy debt.

However, critics point to the "rebound effect" and "enabled emissions" – the carbon emissions facilitated by AI. Holly and Will Alpine, former Microsoft employees who worked on responsible AI and sustainability, resigned due to their employer's partnerships with oil and gas clients. They argue that Microsoft's work with ExxonMobil and Chevron, which aids fossil fuel extraction, enables far more emissions than the company's operational footprint produces.

The IEA estimates AI could boost technically recoverable oil and gas reserves by 5% and cut deepwater offshore project costs by 10%. Mike Sommers, head of the American Petroleum Institute, has likened AI to "the next fracking boom." Saudi Aramco CEO Amin Nasser stated the company has embedded AI "in everything," doubling technology spending to increase productivity and the number of wells. While the industry claims AI can cut its carbon intensity by spotting methane leaks, campaigners like Wilson argue this serves as an excuse to delay action, noting "watching methane from space is not stopping methane."

Beyond production, AI's impact on consumption is also a concern. Studies show generative AI advertisements outperform human-created ones, reducing the cost of encouraging consumption. The marketing industry anticipates AI agents booking holidays and purchasing on behalf of customers, further driving demand.

Calls for intervention are growing. In October, the UN's special rapporteur on the human right to safe drinking water advocated for a moratorium on new datacentre development due to environmental impacts. A coalition of over 230 US environmental groups demanded a national moratorium in December until regulations are established. Ireland's utility commission has since lifted its de facto ban on grid connections, but now requires 80% of a datacentre's annual electricity consumption to eventually come from new renewable projects.

Other proposals include taxing AI to fund climate action, as suggested by Paris Agreement architect Laurence Tubiana. The Alpines advocate for the EU's upcoming AI bill to classify fossil fuel applications of AI as high-risk and for investors to consider enabled emissions in ESG metrics.

While Google and xAI did not respond to requests for comment, OpenAI stated it prioritizes the best use of its computing power, supports sustainability goals, and believes AI will be instrumental in addressing climate change. Microsoft acknowledged the complexity of the energy transition, emphasizing technology's role in decarbonization while balancing present and future energy needs.

Luccioni contends that instead of panic, the focus should be on pressuring companies to design "frugal" AI tools. "I still believe that AI can do good in terms of fighting the climate crisis – designing the next generation of batteries, tracking deforestation, predicting hurricanes," she said, contrasting this potential



with the current reality of "AI slop" on social media powered by diesel generators.

